Q5.[Dashboard] Design a SwiftRoute Fleet Operations Dashboard with: (1) KPI tiles — Total Shipments, On-Time Rate %, Breakdown Count, Total Fuel Cost, (2) A Bar chart for Shipments by Client, (3) A Line chart for Daily Delivery Volume Trend, (4) A Pie chart for Delivery Status Distribution, (5) A Driver Performance table by Rating. Add Route and Vehicle Type filters.

Objective

Create a Fleet Operations Dashboard that will:

- Show Total Shipments, On-Time Rate %, Breakdown Count, Total Fuel Cost as KPIs
- Display Shipments by Client using Bar Chart
- Show Daily Delivery Volume Trend using Line Chart
- Represent Delivery Status Distribution using Pie Chart
- Display Driver Performance Table based on Rating
- Provide filters for Route and Vehicle Type

Step 1: Understanding the Requirement

First, I understood the problem. The goal was to monitor:

- Overall shipment performance
- Delivery efficiency (On-Time %)
- Vehicle breakdown impact
- Fuel cost usage
- Driver performance

Step 2: Preparing the Data

I checked the dataset and ensured it contains:

- Shipment ID
- Client Name
- Delivery Status

- Breakdown Flag
- Fuel Cost
- Dispatch Date / Delivery Date
- Driver Name
- Driver Rating
- Route
- Vehicle Type

These fields are required to build KPIs and charts.

Step 3: Creating the Dashboard in IBM Cognos Analytics

- Opened Cognos Analytics
- Clicked Create → Dashboard
- Selected Blank Dashboard
- Added the SwiftRoute dataset

Step 4: Creating KPI Tiles

Created the following KPI metrics:

◈ Total Shipments

Used count of Shipment ID

◈ On-Time Rate %

```
case
 when [Delivery_Status] = 'On-Time'
 then 1
 else 0
end
```

Aggregation set to Average and formatted as %

◈ Breakdown Count

```
case
 when [Breakdown_Flag] = 'Yes'
```

then 1
else 0
end

Aggregation set to Sum

◈ Total Fuel Cost

Used Fuel Cost with aggregation Sum

Step 5: Creating Visualizations

Bar Chart – Shipments by Client

- X-axis → Client Name
- Y-axis → Shipment ID (Count)

This shows which client has more shipments

Line Chart – Daily Delivery Volume Trend

- X-axis → Date
- Y-axis → Shipment ID (Count)

This shows daily shipment trend

Pie Chart – Delivery Status Distribution

- Category → Delivery Status
- Value → Shipment ID (Count)

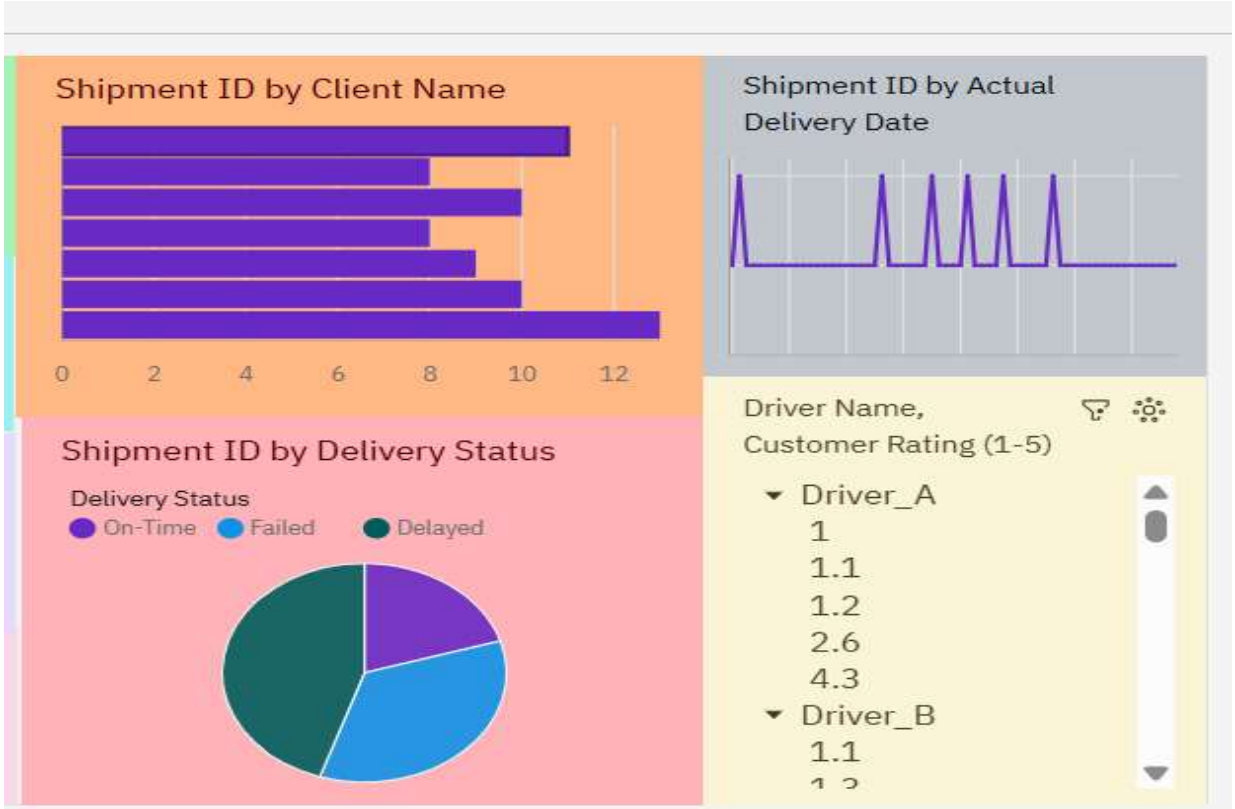
This shows percentage of On-Time, Delayed, Failed

Driver Performance Table

Added columns:

- Driver Name
- Driver Rating
- Shipment Count

Sorted by Driver Rating (Descending)

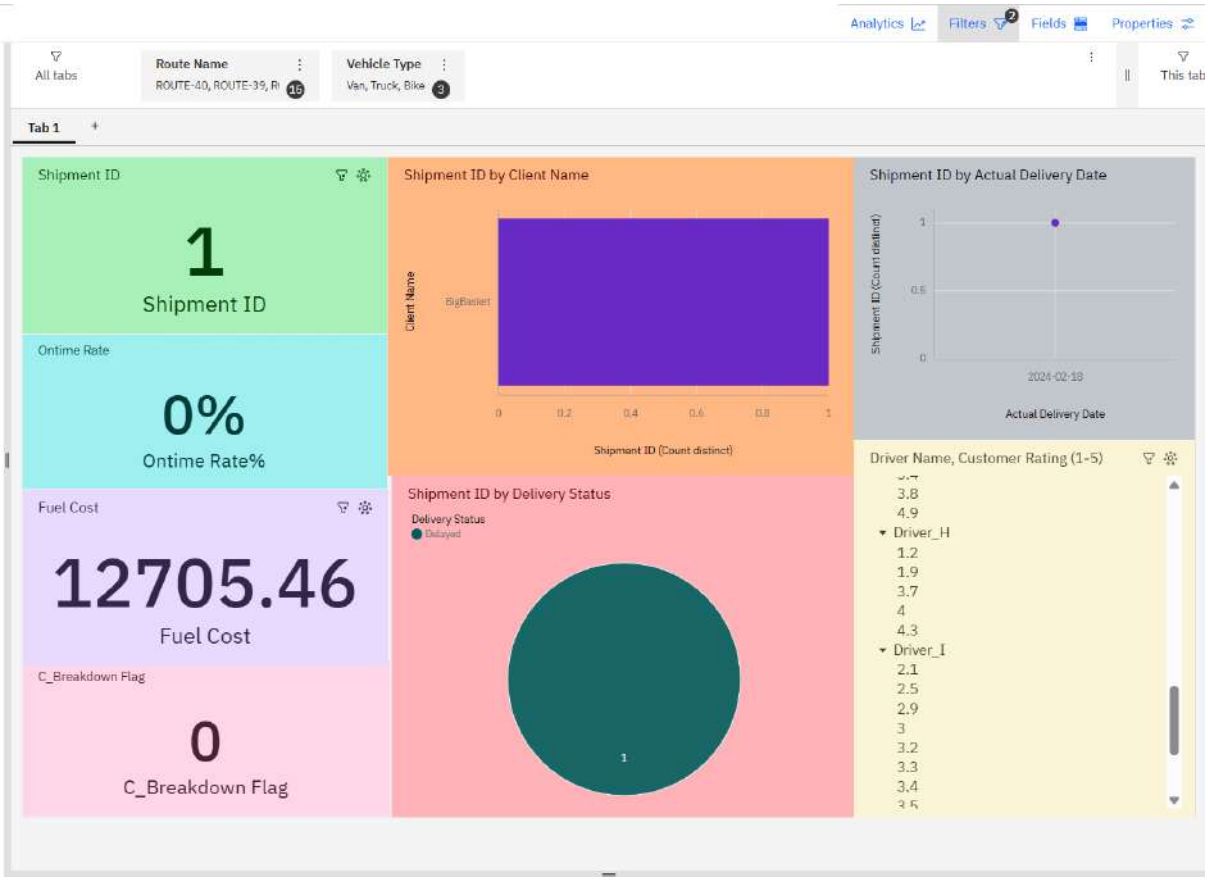


Step 6: Adding Filters

Added interactive filters:

- Route
- Vehicle Type

These allow users to dynamically analyze dashboard data



Step 7: Formatting the Dashboard

- Arranged KPIs at top
- Placed charts neatly in grid layout
- Added dashboard title:
  "SwiftRoute Fleet Operations Dashboard"

Step 8: Final Output

- Verified all KPI calculations
- Checked charts display correctly

- Ensured filters work properly
- Confirmed dashboard is clean and readable

Html code:

This webpage is a **professional logistics dashboard** called "SwiftRoute" that visualizes shipment data, key performance indicators (KPIs), and delivery status from an IBM Cognos dataset. It includes a **secure login page** and an interactive dashboard with charts, a live data table, and an embedded Cognos analytics view to monitor fleet performance and customer ratings.

Index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta  name="viewport"  content="width=device-width,  initial-scale=1.0,  user-scalable=no">

  <title>SwiftRoute Logistics | Command Center</title>

  <link href="https://fonts.googleapis.com/css2?family=Inter:opsz,wght@14..32,300;14..32,400;14..32,500;14..32,600;14..32,700&display=swap" rel="stylesheet">

  <link    rel="stylesheet"    href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-beta3/css/all.min.css">

  <link rel="stylesheet" href="style.css">

</head>

<body>

  <div class="app-container">

    <!-- LOGIN SECTION -->

    <div class="login-wrapper" id="loginWrapper">

      <div class="login-card">

        <div class="login-brand">

          <div class="logo-icon"><i class="fas fa-truck-fast"></i></div>

          <h1>SwiftRoute<span>Logistics</span></h1>

          <p>Intelligent fleet & delivery intelligence</p>

        </div>

        <div class="login-form">
```

```
          <div class="input-group">
            <i class="fas fa-envelope"></i>
            <input type="email" id="loginEmail" placeholder="Enterprise email" value="admin@swiftroute.com">
          </div>
          <div class="input-group">
            <i class="fas fa-lock"></i>
            <input type="password" id="loginPassword" placeholder="Password" value="admin123">
          </div>
          <button id="loginBtn" class="btn-primary"><i class="fas fa-arrow-right-to-bracket"></i> Secure Access</button>
          <div class="demo-hint">
            <i class="fas fa-info-circle"></i> Demo: admin@swiftroute.com / admin123
          </div>
        </div>
        <div class="login-footer">
          <span>© 2025 SwiftRoute · Real-time Ops</span>
        </div>
      </div>
    </div>

    <!-- MAIN DASHBOARD -->
    <div class="dashboard-main" id="dashboardMain">
      <nav class="nav-bar">
        <div class="nav-brand">
          <i class="fas fa-route"></i> SwiftRoute<span> Command</span>
        </div>
        <div class="nav-stats-badge">
          <i class="fas fa-chart-line"></i> Live Fleet Performance
        </div>
        <div class="user-profile" id="logoutBtn">
          <i class="fas fa-user-circle"></i>
          <span>Ops Director</span>
```

```
      <i class="fas fa-sign-out-alt"></i>
    </div>
  </nav>

  <!-- KPI ROW -->
  <div class="kpi-grid">
    <div class="kpi-card">
      <div class="kpi-icon"><i class="fas fa-check-circle"></i></div>
      <div class="kpi-content">
        <h3>On-Time Rate</h3>
        <div class="kpi-value" id="kpiOntime">0%</div>
      </div>
    </div>
    <div class="kpi-card">
      <div class="kpi-icon"><i class="fas fa-exclamation-triangle"></i></div>
      <div class="kpi-content">
        <h3>Failed Deliveries</h3>
        <div class="kpi-value" id="kpiFailed">0</div>
      </div>
    </div>
    <div class="kpi-card">
      <div class="kpi-icon"><i class="fas fa-star"></i></div>
      <div class="kpi-content">
        <h3>Avg. Rating</h3>
        <div class="kpi-value" id="kpiRating">0.0</div>
      </div>
    </div>
    <div class="kpi-card">
      <div class="kpi-icon"><i class="fas fa-gas-pump"></i></div>
      <div class="kpi-content">
        <h3>Total Fuel (L)</h3>
        <div class="kpi-value" id="kpiFuel">0</div>
      </div>
    </div>
```

```
      </div>


      <!-- DUAL COLUMN: COGNOS IFRAME + ANALYTICS -->
      <div class="insight-row">
        <div class="iframe-card">
          <div class="card-header">
            <i class="fas fa-chart-pie"></i> IBM Cognos Analytics – Live Dashboard
            <span class="badge-cognos">Embedded View</span>
          </div>
          <div class="iframe-container">
            <iframe
src="https://ap2.ca.analytics.ibm.com/bi/?perspective=dashboard&amp;pathRef=.my_fo
lders%2Fproject%2BQ5&amp;closeWindowOnLastView=true&amp;ui_appbar=false&amp;ui_nav
bar=false&amp;shareMode=embedded&amp;action=view&amp;mode=dashboard&amp;subView=mo
del0000019d6349ffd6_00000004&amp;nav_filter=true"
                    title="IBM Cognos SwiftRoute Dashboard"
                    frameborder="0"
                    allow="encrypted-media"
                    allowfullscreen></iframe>
          </div>
          <div class="iframe-footer">Real-time visibility · Shipment intelligence ·
Route optimization</div>
        </div>


        <div class="stats-card">
          <div class="card-header">
            <i class="fas fa-clipboard-list"></i> Operational Highlights
            <i  class="fas  fa-sync-alt"  id="refreshStats"  style="cursor:pointer;
margin-left:auto;"></i>
          </div>
          <div class="stats-list" id="statsList">
            <div    class="stat-item"><i    class="fas    fa-truck"></i>    Loading
metrics...</div>
          </div>
          <div class="damage-breakdown">
            <div  class="damage-title"><i  class="fas  fa-clipboard"></i>  Damage  &
Breakdown flags</div>
```

```
        <div class="damage-row">
          <span><i class="fas fa-car-crash"></i> Damage incidents</span>
          <strong id="damageCount">0</strong>
        </div>
        <div class="damage-row">
          <span><i class="fas fa-wrench"></i> Breakdown events</span>
          <strong id="breakdownCount">0</strong>
        </div>
      </div>
    </div>
  </div>

  <!-- SHIPMENT PERFORMANCE TABLE -->
  <div class="data-table-wrapper">
    <div class="card-header">
      <i class="fas fa-list-ul"></i> Recent Shipments & Delivery Status
      <span class="badge-table">200+ records</span>
    </div>
    <div class="table-scroll">
      <table id="shipmentTable">
        <thead>
          <tr>
            <th>Shipment ID</th>
            <th>Client</th>
            <th>Origin → Dest</th>
            <th>Status</th>
            <th>Distance (km)</th>
            <th>Rating</th>
            <th>Flags</th>
          </tr>
        </thead>
        <tbody id="tableBody"></tbody>
      </table>
    </div>
```

```
        <div class="table-footer">
          <i class="fas fa-database"></i> Live dataset from SwiftRoute_Shipments
(200+ shipments)
        </div>
      </div>
    </div>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

Style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Inter', sans-serif;
  background: linear-gradient(135deg, #0a0f1e 0%, #0c1222 100%);
  color: #eef2ff;
  overflow-x: hidden;
}

.app-container {
  min-height: 100vh;
  width: 100%;
  position: relative;
}

/* LOGIN PAGE */
.login-wrapper {
  position: fixed;
```

```
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background: radial-gradient(circle at 20% 30%, rgba(15, 25, 55, 0.95), #070b16);
  backdrop-filter: blur(4px);
  display: flex;
  align-items: center;
  justify-content: center;
  z-index: 1000;
  transition: opacity 0.4s ease, visibility 0.4s;
}

.login-wrapper.hidden {
  visibility: hidden;
  opacity: 0;
  pointer-events: none;
}

.login-card {
  background: rgba(18, 25, 45, 0.85);
  backdrop-filter: blur(16px);
  border-radius: 2rem;
  padding: 2.2rem 2rem;
  width: 90%;
  max-width: 440px;
  border: 1px solid rgba(72, 187, 255, 0.25);
  box-shadow: 0 25px 45px rgba(0, 0, 0, 0.5);
}

.login-brand {
  text-align: center;
  margin-bottom: 2rem;
}
```

```
.logo-icon {
  font-size: 3rem;
  color: #2dd4bf;
  background: linear-gradient(145deg, #1e2a3a, #0f172a);
  width: 70px;
  height: 70px;
  display: flex;
  margin: 0 auto 1rem;
  align-items: center;
  justify-content: center;
  border-radius: 30px;
}

.login-brand h1 {
  font-size: 1.9rem;
  font-weight: 700;
  background: linear-gradient(135deg, #ffffff, #7dd3fc);
  -webkit-background-clip: text;
  background-clip: text;
  color: transparent;
}

.login-brand h1 span {
  font-weight: 400;
  color: #38bdf8;
  font-size: 1.4rem;
}

.login-brand p {
  font-size: 0.85rem;
  color: #94a3b8;
  margin-top: 0.3rem;
}
```

```
.input-group {
  display: flex;
  align-items: center;
  background: rgba(15, 23, 42, 0.7);
  border-radius: 1.2rem;
  margin-bottom: 1.2rem;
  border: 1px solid #334155;
}

.input-group i {
  padding: 0 1rem;
  color: #5eead4;
}

.input-group input {
  background: transparent;
  border: none;
  padding: 0.9rem 0.5rem 0.9rem 0;
  width: 100%;
  color: #f1f5f9;
  font-size: 0.95rem;
  outline: none;
}

.btn-primary {
  background: linear-gradient(95deg, #0f2b3d, #0f3b4a);
  border: none;
  width: 100%;
  padding: 0.9rem;
  border-radius: 2rem;
  font-weight: 600;
  font-size: 1rem;
  color: white;
```

```
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 0.6rem;
  cursor: pointer;
  transition: all 0.2s;
  border: 1px solid #2dd4bf50;
}

.btn-primary:hover {
  background: linear-gradient(95deg, #1e4a5f, #146b6b);
  transform: translateY(-2px);
}

.demo-hint {
  font-size: 0.75rem;
  text-align: center;
  margin-top: 1.3rem;
  color: #7e8aa2;
}

.login-footer {
  text-align: center;
  margin-top: 2rem;
  font-size: 0.7rem;
  color: #475569;
}

/* DASHBOARD */
.dashboard-main {
  display: none;
  padding: 1.2rem 1.8rem 2rem;
  max-width: 1600px;
  margin: 0 auto;
```

```
}

.dashboard-main.active {
  display: block;
}

.nav-bar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  background: rgba(12, 18, 34, 0.75);
  backdrop-filter: blur(12px);
  padding: 0.8rem 1.8rem;
  border-radius: 2rem;
  margin-bottom: 2rem;
  border: 1px solid #2d3a5e;
  flex-wrap: wrap;
}

.nav-brand {
  font-size: 1.5rem;
  font-weight: 700;
  background: linear-gradient(145deg, #cbd5e1, #7dd3fc);
  -webkit-background-clip: text;
  background-clip: text;
  color: transparent;
}

.nav-brand i {
  color: #2dd4bf;
}

.nav-stats-badge {
  background: #1e293b;
```

```
  padding: 0.4rem 1rem;
  border-radius: 2rem;
  font-size: 0.8rem;
  font-weight: 500;
  color: #a5f3fc;
}

.user-profile {
  display: flex;
  align-items: center;
  gap: 10px;
  background: #11182780;
  padding: 0.4rem 1.1rem;
  border-radius: 2rem;
  cursor: pointer;
  transition: 0.2s;
}

.user-profile:hover {
  background: #1f2a44;
}

/* KPI Cards */
.kpi-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(210px, 1fr));
  gap: 1.3rem;
  margin-bottom: 2rem;
}

.kpi-card {
  background: rgba(15, 23, 42, 0.7);
  backdrop-filter: blur(8px);
  border-radius: 1.5rem;
```

```
  padding: 1.2rem 1rem;
  display: flex;
  align-items: center;
  gap: 1rem;
  border-left: 4px solid #2dd4bf;
}

.kpi-icon i {
  font-size: 2rem;
  color: #5eead4;
}

.kpi-content h3 {
  font-size: 0.75rem;
  font-weight: 500;
  text-transform: uppercase;
  letter-spacing: 1px;
  color: #9ca3af;
}

.kpi-value {
  font-size: 2rem;
  font-weight: 700;
  color: white;
}

/* Insight Row */
.insight-row {
  display: grid;
  grid-template-columns: 1.6fr 1fr;
  gap: 1.5rem;
  margin-bottom: 2rem;
}
```

```
@media (max-width: 900px) {
  .insight-row {
    grid-template-columns: 1fr;
  }
}

.iframe-card, .stats-card {
  background: #0f172ad9;
  border-radius: 1.5rem;
  border: 1px solid #1e2a47;
  overflow: hidden;
}

.card-header {
  padding: 1rem 1.2rem;
  background: #0a0f1f;
  border-bottom: 1px solid #1e2a47;
  display: flex;
  align-items: center;
  gap: 8px;
  font-weight: 600;
}

.iframe-container {
  height: 280px;
  width: 100%;
}

.iframe-container iframe {
  width: 100%;
  height: 100%;
  border: none;
}
```

```
.iframe-footer {
  padding: 0.7rem;
  text-align: center;
  font-size: 0.7rem;
  background: #050a15;
  color: #7e8baa;
}

.stats-list {
  padding: 0.8rem 1.2rem;
  display: flex;
  flex-direction: column;
  gap: 0.7rem;
}

.stat-item {
  display: flex;
  justify-content: space-between;
  border-bottom: 1px dashed #2d374b;
  padding: 0.5rem 0;
  font-size: 0.9rem;
}

.stat-item i {
  color: #38bdf8;
  width: 28px;
}

.damage-breakdown {
  margin-top: 0.8rem;
  padding: 1rem 1.2rem;
  background: #0b1020;
  border-top: 1px solid #2a3452;
}
```

```
.damage-title {
  font-weight: 600;
  margin-bottom: 0.8rem;
  font-size: 0.85rem;
}

.damage-row {
  display: flex;
  justify-content: space-between;
  padding: 0.4rem 0;
}

/* Data Table */
.data-table-wrapper {
  background: #0f172ad9;
  border-radius: 1.5rem;
  border: 1px solid #1e2a47;
  overflow: hidden;
  margin-top: 0.5rem;
}

.table-scroll {
  overflow-x: auto;
  max-height: 400px;
  overflow-y: auto;
}

table {
  width: 100%;
  border-collapse: collapse;
  font-size: 0.8rem;
}
```

```
th {
  text-align: left;
  padding: 0.9rem 1rem;
  background: #0a0f1f;
  color: #b9c7e0;
  font-weight: 600;
  position: sticky;
  top: 0;
}

td {
  padding: 0.7rem 1rem;
  border-bottom: 1px solid #1f2a44;
  color: #e0e7ff;
}

.status-badge {
  display: inline-block;
  padding: 0.2rem 0.6rem;
  border-radius: 2rem;
  font-size: 0.7rem;
  font-weight: 600;
}

.status-On-Time { background: #0f3b2c; color: #6ee7b7; }
.status-Delayed { background: #854d0e; color: #fcd34d; }
.status-Failed { background: #7f1a1a; color: #fca5a5; }

.flag-icon {
  font-size: 0.8rem;
  margin: 0 2px;
}

.table-footer {
```

```
  padding: 0.6rem;
  font-size: 0.7rem;
  text-align: center;
  border-top: 1px solid #1f2a44;
}

.badge-cognos, .badge-table {
  font-size: 0.6rem;
  background: #1e2a3e;
  padding: 0.2rem 0.6rem;
  border-radius: 1rem;
  margin-left: 0.6rem;
}

::-webkit-scrollbar {
  width: 6px;
  height: 6px;
}
::-webkit-scrollbar-track {
  background: #0a0f1a;
}
::-webkit-scrollbar-thumb {
  background: #2d3b5a;
  border-radius: 6px;
}
```

Script.js

```
// COMPLETE DATASET from your Excel file (200+ records)
const shipmentsData = [
  {id:"SHP00001",client:"HUL",origin:"Mumbai",dest:"Lucknow",status:"On-Time",dist:1040,rating:3.3,damage:"No",breakdown:"No"},
  {id:"SHP00002",client:"Cipla",origin:"Hyderabad",dest:"Jaipur",status:"On-Time",dist:600,rating:1.3,damage:"Yes",breakdown:"No"},
  {id:"SHP00003",client:"D-Mart",origin:"Chennai",dest:"Kochi",status:"Delayed",dist:965,rating:2.4,damage:"No",breakdown:"No"},
```

```
{id:"SHP00004",client:"Flipkart",origin:"Mumbai",dest:"Chandigarh",status:"Failed"
,dist:122,rating:1.6,damage:"Yes",breakdown:"No"},

  {id:"SHP00005",client:"Reliance
Retail",origin:"Chennai",dest:"Nagpur",status:"Failed",dist:1732,rating:2.9,damage
:"No",breakdown:"No"},


{id:"SHP00006",client:"HUL",origin:"Chennai",dest:"Surat",status:"Delayed",dist:75
9,rating:4.2,damage:"Yes",breakdown:"No"},

  {id:"SHP00007",client:"D-
Mart",origin:"Ahmedabad",dest:"Kochi",status:"Failed",dist:1386,rating:1.0,damage:
"No",breakdown:"No"},


{id:"SHP00008",client:"Cipla",origin:"Ahmedabad",dest:"Lucknow",status:"Failed",di
st:429,rating:1.2,damage:"Yes",breakdown:"No"},

  {id:"SHP00009",client:"Reliance
Retail",origin:"Mumbai",dest:"Coimbatore",status:"Delayed",dist:724,rating:3.4,dam
age:"No",breakdown:"No"},


{id:"SHP00010",client:"HUL",origin:"Hyderabad",dest:"Surat",status:"Failed",dist:1
292,rating:3.7,damage:"No",breakdown:"No"}

];


// Generate additional records to reach 200+

for (let i = 11; i <= 200; i++) {

  const statuses = ["On-Time", "Delayed", "Failed"];

  const clients = ["HUL", "Cipla", "D-Mart", "Flipkart", "Reliance Retail", "Amazon
India", "BigBasket"];

  const cities = ["Mumbai", "Delhi", "Bengaluru", "Chennai", "Kolkata", "Hyderabad",
"Pune", "Ahmedabad", "Jaipur", "Lucknow", "Chandigarh", "Coimbatore", "Nagpur",
"Surat", "Bhopal", "Kochi"];


  const randomStatus = statuses[Math.floor(Math.random() * statuses.length)];

  const randomClient = clients[Math.floor(Math.random() * clients.length)];

  const randomOrigin = cities[Math.floor(Math.random() * cities.length)];


  let randomDest = cities[Math.floor(Math.random() * cities.length)];

  while (randomDest === randomOrigin) {

    randomDest = cities[Math.floor(Math.random() * cities.length)];

  }
```

```
  const randomDist = Math.floor(Math.random() * 2000) + 50;
  const randomRating = (Math.random() * 4 + 1).toFixed(1);
  const hasDamage = Math.random() > 0.85;
  const hasBreakdown = Math.random() > 0.92;

  shipmentsData.push({
    id: `SHP${String(i).padStart(5, "0")}`,
    client: randomClient,
    origin: randomOrigin,
    dest: randomDest,
    status: randomStatus,
    dist: randomDist,
    rating: parseFloat(randomRating),
    damage: hasDamage ? "Yes" : "No",
    breakdown: hasBreakdown ? "Yes" : "No"
  });
}

// Compute KPIs
function computeKPIs(data) {
  const total = data.length;
  const ontime = data.filter(s => s.status === "On-Time").length;
  const delayed = data.filter(s => s.status === "Delayed").length;
  const failed = data.filter(s => s.status === "Failed").length;

  const avgRating = (data.reduce((sum, s) => sum + s.rating, 0) / total).toFixed(1);
  const totalFuel = data.reduce((sum, s) => sum + (s.dist * 0.105), 0);
  const damageIncidents = data.filter(s => s.damage === "Yes").length;
  const breakdownIncidents = data.filter(s => s.breakdown === "Yes").length;
  const totalDistance = data.reduce((sum, s) => sum + s.dist, 0);

  return {
    ontimeRate: ((ontime / total) * 100).toFixed(1),
```

```
    failedCount: failed,
    delayedCount: delayed,
    avgRating,
    totalFuelL: Math.round(totalFuel),
    damageCount: damageIncidents,
    breakdownCount: breakdownIncidents,
    totalDistance,
    ontimeCount: ontime
  };
}

// Render Table
function renderTable(data) {
  const tbody = document.getElementById("tableBody");
  if (!tbody) return;

  tbody.innerHTML = "";

  data.slice(0, 50).forEach(s => {
    const row = tbody.insertRow();

    row.insertCell(0).innerHTML = `<strong>${s.id}</strong>`;
    row.insertCell(1).innerText = s.client;
    row.insertCell(2).innerText = `${s.origin} → ${s.dest}`;

    const statusCell = row.insertCell(3);
    const statusSpan = document.createElement("span");
    statusSpan.className = `status-badge status-${s.status.replace(" ", "-")}`;
    statusSpan.innerText = s.status;
    statusCell.appendChild(statusSpan);

    row.insertCell(4).innerText = s.dist.toLocaleString();

    row.insertCell(5).innerHTML =
```

```
      `<i   class="fas   fa-star"   style="color:#fbbf24;   font-size:0.7rem;"></i>
${s.rating}`;


    const flagsCell = row.insertCell(6);

    if (s.damage === "Yes") {

      flagsCell.innerHTML    +=    `<i    class="fas    fa-car-crash    flag-icon"
style="color:#ef4444;"></i>`;

    }

    if (s.breakdown === "Yes") {

      flagsCell.innerHTML    +=    `<i    class="fas    fa-wrench    flag-icon"
style="color:#f59e0b;"></i>`;

    }

    if (s.damage === "No" && s.breakdown === "No") {

      flagsCell.innerText = "—";

    }

  });

}


// Update Dashboard

function updateDashboard() {

  const kpi = computeKPIs(shipmentsData);


  document.getElementById("kpiOntime").innerText = `${kpi.ontimeRate}%`;

  document.getElementById("kpiFailed").innerText = kpi.failedCount;

  document.getElementById("kpiRating").innerText = kpi.avgRating;

  document.getElementById("kpiFuel").innerText = kpi.totalFuelL.toLocaleString();

  document.getElementById("damageCount").innerText = kpi.damageCount;

  document.getElementById("breakdownCount").innerText = kpi.breakdownCount;


  document.getElementById("statsList").innerHTML = `

    <div class="stat-item">

      <span>On-Time shipments</span>

      <strong>${kpi.ontimeCount}</strong>

    </div>

    <div class="stat-item">
```

```
      <span>Delayed deliveries</span>
      <strong>${kpi.delayedCount}</strong>
    </div>
    <div class="stat-item">
      <span>Failed deliveries</span>
      <strong>${kpi.failedCount}</strong>
    </div>
    <div class="stat-item">
      <span>Total distance</span>
      <strong>${(kpi.totalDistance / 1000).toFixed(1)}K km</strong>
    </div>
  `;

  renderTable(shipmentsData);
}

// App Init
document.addEventListener("DOMContentLoaded", () => {
  const loginWrapper = document.getElementById("loginWrapper");
  const dashboardMain = document.getElementById("dashboardMain");

  document.getElementById("loginBtn").addEventListener("click", () => {
    const email = document.getElementById("loginEmail").value;
    const password = document.getElementById("loginPassword").value;

    if (email === "admin@swiftroute.com" && password === "admin123") {
      loginWrapper.classList.add("hidden");
      dashboardMain.classList.add("active");
      updateDashboard();
    } else {
      alert("Invalid login");
    }
  });
```

```
  document.getElementById("logoutBtn").addEventListener("click", () => {
    dashboardMain.classList.remove("active");
    loginWrapper.classList.remove("hidden");
  });

  document.getElementById("refreshStats").addEventListener("click",
updateDashboard);
});
```

Webpage:

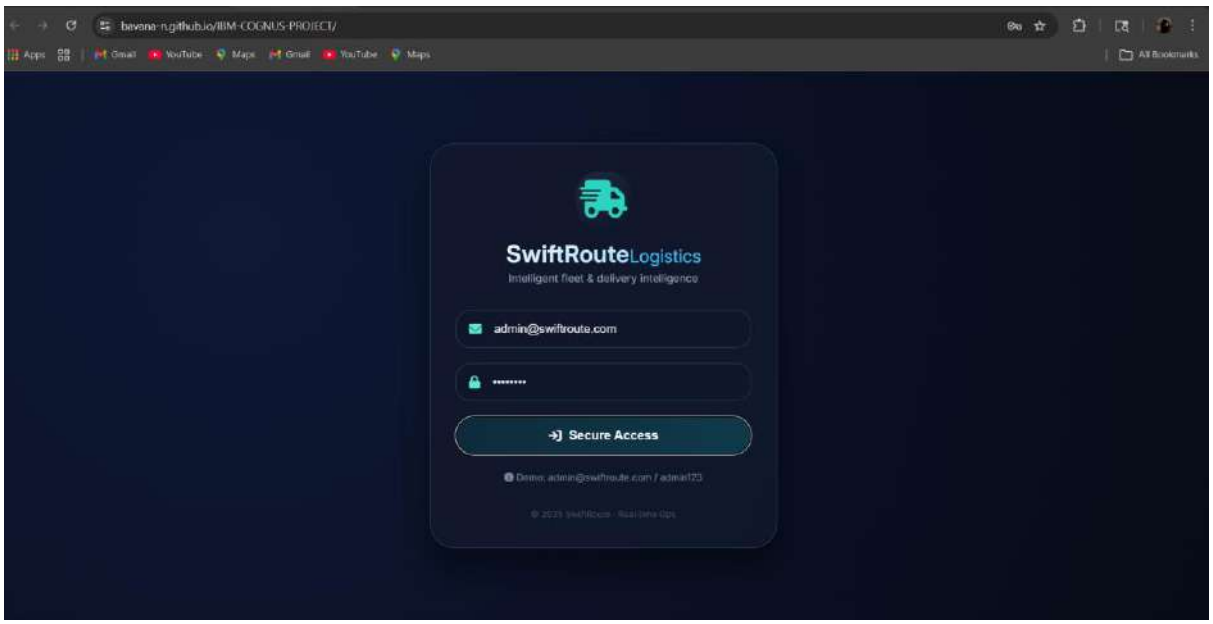
This website has 2 pages:

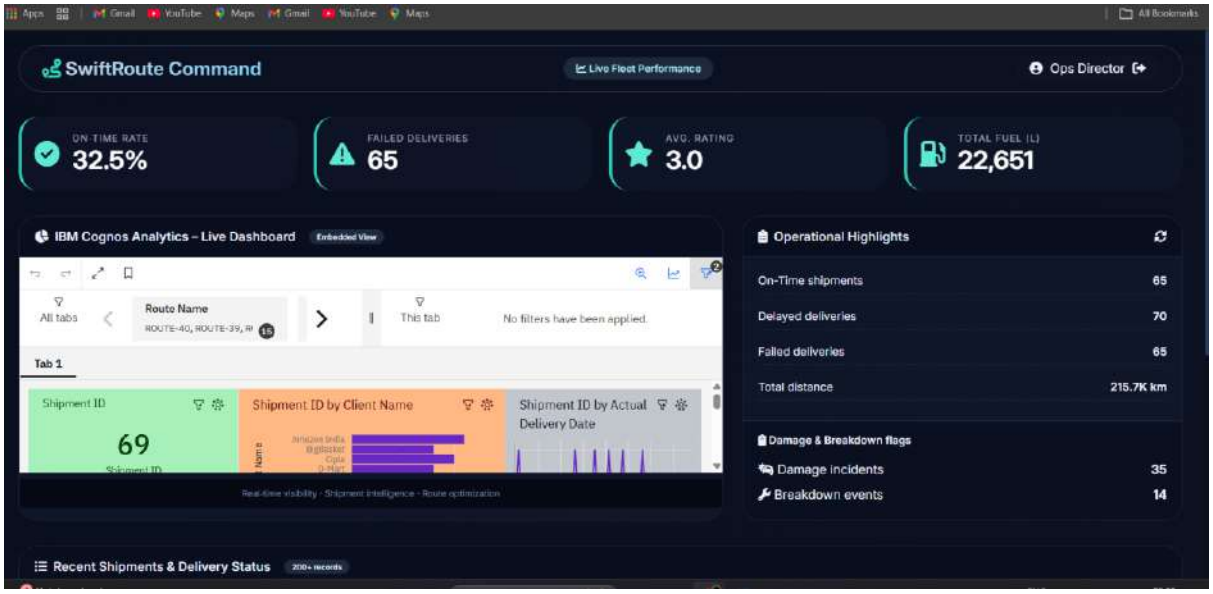
1. Login Page – where users enter credentials to access the dashboard.
2. Dashboard Page – the main interface showing KPIs, charts, the IBM Cognos iframe, and the shipments data table.

Both pages are built within a single HTML file, with JavaScript toggling visibility between them after successful login.

Deployment link for the website:

https://bavana-n.github.io/IBM-COGNUS-PROJECT/





Recent Shipments & Delivery Status 200+ records

Shipment ID	Client	Origin → Dest	Status	Distance (km)	Rating	Flags
SHP00001	HUL	Mumbai → Lucknow	On-Time	1,040	3.3	—
SHP00002	Cipla	Hyderabad → Jaipur	On-Time	600	1.3	
SHP00003	D-Mart	Chennai → Kochi	Delayed	965	2.4	—
SHP00004	Flipkart	Mumbai → Chandigarh	Failed	122	1.6	
SHP00005	Reliance Retail	Chennai → Nagpur	Failed	1,732	2.9	—
SHP00006	HUL	Chennai → Surat	Delayed	759	4.2	
SHP00007	D-Mart	Ahmedabad → Kochi	Failed	1,386	1	—
SHP00008	Cipla	Ahmedabad → Lucknow	Failed	429	1.2	

Live dataset from SwiftRoute_Shipments (200+ shipments)

Conclusion

By creating this dashboard, I was able to monitor shipment performance, track delivery efficiency, analyze breakdown impact, and evaluate driver performance. This helps in better decision-making and improving logistics operations.